

Multi-Agent Scientific Paper Review

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File	Final_paper_kjns_210426 (1).pdf
Words	12,362
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	DEEP
Model	llama3.2

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Editorial Decision

MINOR REVISION

Weighted Score: 3.8 / 5.0
Confidence: 75%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	4.2	Good	■■■■■
Manuscript Structure	4.0	Good	■■■■■
Methodology & Equations	4.2	Good	■■■■■
Statistical Analysis	3.5	Good	■■■■■
Results & Evidence	4.0	Good	■■■■■
Discussion & Conclusions	4.2	Good	■■■■■
Figures & Tables	4.0	Good	■■■■■
Scientific Writing	2.5	Weak	■■■■■
References & Citations	2.5	Weak	■■■■■
Ethics & Transparency	3.5	Good	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 4.2 / 5.0 | Confidence: 80%

Strengths:

- [TITLE] The title clearly signals the core method and application domain.
- [ABSTRACT] The abstract provides substantial methodological and validation detail.
- [ABSTRACT] Quantitative result signals are present in the abstract.
- [KEYWORDS] 8 keywords were detected for indexing and discoverability.

Weaknesses:

- [TITLE] could be shortened to fit within the recommended length.

Major Comments:

- [ABSTRACT] could benefit from a clearer structure and more concise language.

StructureReviewer

Manuscript structure and logical coherence

Score: 4.0 / 5.0 | Confidence: 80%

Strengths:

- The manuscript presents a clear research problem and explicitly declares the research question.
- The structure flow is generally logical, with each section flowing into the next.

Weaknesses:

- Objectives are stated but not clearly addressed in the conclusions.
- Some sections (e.g., Section VI) seem to be duplicates of Section V.

Recommendations:

- Clarify how objectives are addressed in the conclusions.
- Consider removing duplicate sections for better coherence.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: 4.2 / 5.0 | Confidence: 80%

Strengths:

- The approach integrates physical admissibility constraints into the neural mapping.
- Deterministic forward inference is enabled.

Major Comments:

- The logical sequence of the paper is generally good, but there are some gaps between the proposed method and the experiments. The validation directly tests what the method claims to solve, which is a strength.
- The authors could provide more details on the preprocessing steps applied to the measured signals.

Recommendations:

- Consider providing more details on the preprocessing steps applied to the measured signals.
- Verify that the accelerometers were calibrated correctly for each measurement point.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: 3.5 / 5.0 | Confidence: 60%

Strengths:

- Bootstrap confidence intervals, Holm correction, CV, RMSE/MSE, and Pareto indicators are reported as strengths
- The manuscript reports multiple statistical validation elements, including independent runs, nonparametric tests, correction procedures, confidence intervals, and performance/dispersion metrics.

Major Comments:

- The manuscript could benefit from a more detailed discussion on the assumptions of the Kruskal-Wallis test
- It is unclear whether the sample size is sufficient for robustness and significance assessment

Recommendations:

- Consider using non-parametric methods or bootstrapping to verify assumption of normality for Kruskal-Wallis test
 - Provide more detailed justification for the choice of sample size
-

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **4.0 / 5.0** | Confidence: 62%

Strengths:

- 6 figure references were detected in the manuscript.
- 10 table references were detected in the manuscript.
- 52 sequentially numbered equations were detected, suggesting a substantial mathematical formulation.

Weaknesses:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Minor Comments:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Recommendations:

- Manually inspect captions, axes, units, table headers, and variable definitions in the final PDF.
-

ResultsReviewer

Results validity and empirical support

Score: **4.0 / 5.0** | Confidence: 80%

Strengths:

- The manuscript presents a clear research objective and framework for the problem.

Weaknesses:

- Some results, such as Pareto analysis and hypervolume/spread indicators, are not clearly explained or justified in the text.
- The comparison with baseline methods (GA and PSO) is mentioned but not thoroughly discussed.

Major Comments:

- The Results section could benefit from more detailed explanations of the results presented.

Recommendations:

- Provide a clear explanation for each detected result signal, including MSE, RMSE, CV/dispersion, and Pareto analysis.
 - Discuss the comparison with baseline methods (GA and PSO) in more detail, including performance metrics and discussion of results.
-

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **4.2 / 5.0** | Confidence: 80%

Weaknesses:

- The Discussion could benefit from more explicit connections to existing literature on structural parameter identification and multi-objective optimization.

Major Comments:

- The Discussion effectively highlights the strengths of EANN over GA and PSO, but could delve deeper into the theoretical implications of these findings.
- The comparison between methods is mostly based on statistical significance; a more nuanced discussion of practical utility would strengthen the paper.

Recommendations:

- Include more explicit references to relevant literature in the Discussion section.
 - Consider discussing potential limitations and future directions for EANN, such as its applicability to complex systems or scalability issues.
-

WritingReviewer

Scientific writing quality, clarity, and style

Score: **2.5 / 5.0** | Confidence: 80%

Weaknesses:

- Academic tone: formal, objective, third-person where appropriate
- Grammar and syntax correctness

Major Comments:

- The manuscript could benefit from a clearer introduction that sets the stage for the research question.
- The experimental results should be presented in a more detailed manner, including raw data and error analysis.

Recommendations:

- Conciseness: no redundant phrases or unnecessary repetition
 - Paragraph structure: topic sentence + development + transition
-

ReferencesReviewer

References quality, recency, and relevance

Score: **2.5 / 5.0** | Confidence: 90%

Major Comments:

- The references section could benefit from more recent publications, with at least one reference published within the last two years.

Minor Comments:

- The parser detected approximately 34 references; assess recency and relevance from the visible list rather than treating the section as empty.

Recommendations:

- Consider adding more recent references to strengthen the literature review and demonstrate the novelty of the work.
-

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **3.5 / 5.0** | Confidence: 60%

Strengths:

- No clear human/animal-subject signal was detected in the parsed manuscript.

Weaknesses:

- Conflict-of-interest declaration was not detected.

Major Comments:

- Conflict-of-interest declaration was not detected.

Recommendations:

- Include a conflict-of-interest declaration.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] [ABSTRACT] could benefit from a clearer structure and more concise language.
- [2] The logical sequence of the paper is generally good, but there are some gaps between the proposed method and the experiments. The validation directly tests what the method claims to solve, which is a strength.
- [3] The authors could provide more details on the preprocessing steps applied to the measured signals.
- [4] The manuscript could benefit from a more detailed discussion on the assumptions of the Kruskal-Wallis test
- [5] It is unclear whether the sample size is sufficient for robustness and significance assessment
- [6] The Results section could benefit from more detailed explanations of the results presented.
- [7] The Discussion effectively highlights the strengths of EANN over GA and PSO, but could delve deeper into the theoretical implications of these findings.
- [8] The comparison between methods is mostly based on statistical significance; a more nuanced discussion of practical utility would strengthen the paper.
- [9] The manuscript could benefit from a clearer introduction that sets the stage for the research question.
- [10] The experimental results should be presented in a more detailed manner, including raw data and error analysis.
- [11] The references section could benefit from more recent publications, with at least one reference published within the last two years.
- [12] Conflict-of-interest declaration was not detected.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.
- [2] The parser detected approximately 34 references; assess recency and relevance from the visible list rather than treating the section as empty.

Specific Improvement Recommendations

1. Clarify how objectives are addressed in the conclusions.
2. Consider removing duplicate sections for better coherence.
3. Consider providing more details on the preprocessing steps applied to the measured signals.
4. Verify that the accelerometers were calibrated correctly for each measurement point.
5. Consider using non-parametric methods or bootstrapping to verify assumption of normality for Kruskal-Wallis test
6. Provide more detailed justification for the choice of sample size
7. Manually inspect captions, axes, units, table headers, and variable definitions in the final PDF.
8. Provide a clear explanation for each detected result signal, including MSE, RMSE, CV/dispersion, and Pareto analysis.
9. Discuss the comparison with baseline methods (GA and PSO) in more detail, including performance metrics and discussion of results.
10. Include more explicit references to relevant literature in the Discussion section.
11. Consider discussing potential limitations and future directions for EANN, such as its applicability to complex systems or scalability issues.
12. Conciseness: no redundant phrases or unnecessary repetition

This review was generated automatically by the Multi-Agent Paper Review System using Phi-3 Mini (Microsoft) running locally via Ollama. No manuscript content was transmitted to external servers. Rubrics based on Elsevier, IEEE, and Emerald Q1 peer review standards. This document does not constitute an official editorial decision.